

modulated signal is transmitted through the antenna (ANT.).

Using trimmer VC1, tune the carrier frequency in FM band and confirm it with an oscilloscope. You will hear the conversation picked up by MIC1 when you tune frequency of the FM radio set to match frequency of the carrier.

Prepare the coil L1 using about 25cm length of 25SWG

wire. Wrap the wire around a cylindrical object of 6mm diameter and take it out after eight turns.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. After proper soldering, fix switch S1 on the front side of the cabinet. Ensure that the oscillator is properly tuned. Addition of a dipole antenna will extend range of the FM bug transmitter. The circuit works off a 3V battery. ●

Calling Bell Using an Intercom

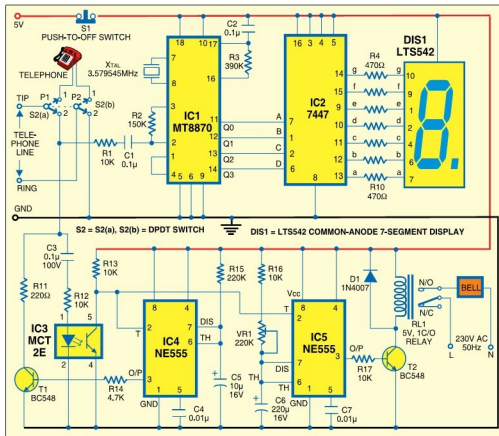
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Here is a simple calling bell circuit that can be used in small offices to call the office boy using an existing intercom system. The office boy can be called from up to nine locations where extension lines are

installed. The system is connected to a dedicated extension for the office boy. Whenever someone needs the office boy's assistance, he can dial the office boy's extension number through the intercom and then press a key to indicate his location number (say, 5). This number will be displayed on a seven-

segment display and at the same time a bell will sound to alert the office boy. Pressing a switch will clear the display.

The entire circuit can be divided into two sections: ring cease cum bell timer, and decoder cum display.



The ring cease and bell timer section is built around optocoupler MCT2E (IC3), NE555 timers (IC4 and IC5) and a few discrete components. The optocoupler is used to trigger the timers (IC4 and IC5). Both the timers are wired in monostable mode. The time period of IC4 is fixed at 2-3 seconds, while the time period of IC5 can be set from 3 seconds to one minute using preset VR1. If you want a longer time period, increase the value of capacitor C6 or resistor R16.

When someone rings the office boy's phone, the internal LED of optocoupler IC3 glows, triggering timers IC4 and IC5. The high output of IC4 at its pin 3 makes transistor T1 conduct, putting resistor R11 across the telephone